

Course Proposal: Hands-On Network Security and System Administration Lab

You can view this course introductory slides at
https://alanshlam.github.io/lab/proposed_course_intro.pdf

Course Description

This course provides a practical, hands-on introduction to the tools and techniques used for modern network security and system administration. Students will learn to deploy, configure, and utilize a wide range of industry-standard open-source tools in realistic lab scenarios. The curriculum covers foundational system and network management skills, progresses to advanced monitoring and security auditing, and introduces key concepts in intrusion prevention and log analysis. By the end of the course, students will have built a comprehensive monitoring and security analysis environment from the ground up, explored several real-world attack vectors, and been introduced to the power of AI in automating and enhancing security and administration tasks.

Learning Outcomes

Upon successful completion of this course, students will be able to:

1. **Apply Foundational Skills:** Demonstrate core system and network administration skills in a Linux environment and set up containerized lab environments with Docker.
2. **Deploy and Configure Monitoring Tools:** Install and configure common network monitoring and security tools, including iperf3, MRTG, ntopng, Suricata, Metasploit, Nmap, the TIG Stack, and the Elastic Stack.
3. **Conduct Security Audits:** Perform network discovery, vulnerability scanning (Nmap), and penetration testing (Metasploit) to identify and understand system weaknesses.
4. **Implement Intrusion Prevention:** Deploy Suricata as an Intrusion Prevention System (IPS), write custom rules, and use iptables to actively block malicious traffic.
5. **Create Monitoring Dashboards:** Build custom dashboards in Grafana and Kibana to visualize time-series data and security events from various sources.

6. **Analyze Common Attack Vectors:** Understand and identify the mechanisms behind common attacks such as DDoS, Man-in-the-Middle (MITM), and Phishing with DNS spoofing.
 7. **Explore AI in Operations:** Investigate the application of AI and Large Language Models (LLMs) for automated vulnerability assessment, forensic investigation, and system/network management tasks.
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Course Syllabus

1. **Week 1: Foundations of the Lab Environment * Topics:** Course overview, introduction to lab environment setup (Cloud vs. Local VM), Linux command-line essentials, fundamental networking concepts (TCP/IP, OSI model), and basic system/network management skills. * **Lab:** System setup, familiarization with the Linux CLI, and basic network troubleshooting commands.
2. **Week 2: Performance Monitoring & Traditional Analysis * Topics:** Docker containerization fundamentals, Simple Network Management Protocol (SNMP), web-based graphing. * **Lab:** Using iperf3 for network performance measurement; configuring and deploying MRTG for basic SNMP graphing.
3. **Week 3: Real-time Traffic Analysis & Advanced Dashboarding * Topics:** Deep packet inspection, time-series databases, data collection agents. * **Lab:** Setting up ntopng for real-time traffic analysis; installing the TIG Stack (Telegraf, InfluxDB, Grafana) for advanced monitoring.
4. **Week 4: Security Audit * Topics:** The penetration testing lifecycle, vulnerability scanning, understanding exploits and payloads. * **Lab:** Using Nmap for network discovery and vulnerability scanning; installing the Metasploit Framework to exploit vulnerabilities on a target.
5. **Week 5: Intrusion Detection and Prevention Systems (IDPS) * Topics:** Principles of IDS vs. IPS, rule-based threat detection, and network traffic redirection with iptables. * **Lab:** Installing and configuring Suricata in IPS mode. Writing custom rules to detect and block attacks discovered in the previous week's audit.
6. **Week 6: Centralized Security Log Management (SIEM) * Topics:** Introduction to the Elastic Stack (ELK) for centralized logging, SIEM concepts. * **Lab:** Installing Elasticsearch, Kibana, and Filebeat. Configuring Filebeat to ship Suricata logs to Elasticsearch and visualizing the results.
7. **Week 7: Real Case Studies in Network Attacks * Topics:** Principles of Denial of Service, Man-in-the-Middle (MITM), and DNS spoofing/phishing attacks. * **Lab:** Simulating a Smurf Attack (ICMP DDoS); demonstrating an

MITM attack with HTTPS interception via ARP Poisoning; executing a DNS spoofing attack.

8. **Week 8: AI in Cybersecurity and System Administration** * **Topics:** The role of machine learning in security, using LLMs for analysis and task automation. * **Lab:** Automatic vulnerability assessment and forensic log investigation with AI tools; using LLMs to generate system administration scripts and suggest network configurations.

Learning Activities

- **Lectures and Laboratory Sessions:** 3 hours per week.

Assessment

- **Lab Assignments (100%):** The final grade is based entirely on the successful completion and submission of weekly lab assignments, which require students to configure the tools and provide evidence of their correct implementation.